

Yes. Incident 01 should now take everything the learner experienced in the Prologue and turn it into disciplined security reasoning.

The learner is **not solving PQ-001 yet**. They are learning how to look at an incident without jumping to conclusions.

Incident 01 — Understand the Incident

1. Core purpose

The Prologue ended with:

PQ-001 OPEN

File 3 attempt 1	NOT RECEIVED
File 3 attempt 2	RECEIVED ALTERED
Alice original	CONFIRMED
Bob copy	DOES NOT MATCH
Credential activity	SUSPICIOUS
Eve authorised test	DOES NOT EXPLAIN IT
Actor	UNKNOWN
Root cause	UNKNOWN

Incident 01 asks:

What can we actually prove happened?

Not:

Who attacked us?

And not yet:

How do we protect the file?

Those come later.

The learner first needs to distinguish:

WHAT WE KNOW
≠
WHAT WE THINK
≠
WHAT WE STILL NEED TO FIND OUT

2. Starting state

The player returns to Central Operations after opening PQ-001.

A notification appears:

CLEARANCE UPDATED

TRAINING & FOUNDATIONS

ACCESS GRANTED ✓

The new objective is:

UNDERSTAND PQ-001

The learner now has access to the evidence gathered in the Prologue, but they still have only basic security tools.

3. What the learner has available

The PQ-001 case contains only evidence already earned.

File evidence

01_LAYOUT.PNG

Delivered normally

02_REPORT.PDF

Delivered normally

03_ACCESS_SCHEDULE.XLSX

Attempt 1: Not received

Attempt 2: Received altered

Alice evidence

SOURCE OWNER

Alice

Original File 3

Confirmed

Bob evidence

RECEIVER

Bob

```
Attempt 1
Nothing received

Attempt 2
File received

Comparison
Does not match Alice original
```

Identity evidence

```
Trainee credential
Created normally

Eve authorised credential test
Completed normally

Later credential event
Unrecognised
```

No attacker name.

No malware.

No IP address yet.

No encryption.

No network attack terminology.

4. Incident 01 structure

We already locked **seven stages**:

1. RECONSTRUCT
2. FACT vs ASSUMPTION
3. TRACE THE EVIDENCE
4. IDENTIFY THE FAILURE
5. IDENTITY / ACCESS
6. TRUST / VERIFICATION
7. FINAL ASSESSMENT

The incident should feel like one investigation moving progressively deeper.

STAGE 1 — RECONSTRUCT

Core question

What happened, and in what order?

The Training & Foundations room contains an interactive incident table.

The evidence from PQ-001 appears as draggable event objects.

For example:

Alice provides File 3

Bob reports no file

File 3 sent

Alice confirms original

File 3 sent again

Bob receives File 3

Bob reports mismatch

Credential anomaly detected

They are initially unordered.

Player action — rebuild the timeline

The learner physically arranges the events.

Correct sequence:

ALICE PROVIDES FILE 3

↓

TRAINEE SENDS FILE

↓

BOB RECEIVES NOTHING

↓

TRAINEE CHECKS WITH ALICE

↓

ALICE CONFIRMS SOURCE

↓

TRAINEE RETRIES

↓
BOB RECEIVES FILE
↓
BOB REPORTS MISMATCH
↓
ALICE CONFIRMS DIFFERENCE
↓
CREDENTIAL ANOMALY REVIEWED

The successful File 1 and File 2 transfers can sit above the timeline as the **normal baseline**.

FILE 1	NORMAL ✓
FILE 2	NORMAL ✓
FILE 3	ABNORMAL ⚠

That contrast matters.

Poor decision behaviour

If the learner creates an impossible sequence, don't say:

Wrong.

Instead:

SEQUENCE CONFLICT

Bob cannot report receiving File 3
before the transfer occurs.

Or:

MISSING EVENT

The retry cannot be evaluated
without confirming the source first.

This teaches causal reasoning.

Stage 1 outcome

TIMELINE
RECONSTRUCTED ✓

NORMAL BASELINE

ESTABLISHED ✓

ANOMALY POINTS

IDENTIFIED ✓

The learner now knows **when** the problem occurred.

They still don't know **why**.

STAGE 2 — FACT vs ASSUMPTION

This is one of the most important Incident 01 mechanics.

Core question

What does the evidence actually support?

The learner gets statements based on PQ-001.

They must place them into:

FACT

ASSUMPTION

UNKNOWN

Examples

Statement

Bob did not receive the first File 3 transfer.

FACT ✓

Because Bob directly reported it.

Statement

The second copy differs from Alice's original.

FACT ✓

Because Alice and Bob's versions were compared.

Statement

Someone read the file.

UNKNOWN

We do not have evidence proving this.

Statement

Eve changed the file.

ASSUMPTION

There is no evidence establishing Eve as responsible.

Statement

An unrecognised credential event occurred.

FACT ✓

Statement

The suspicious credential activity caused the File 3 problem.

UNKNOWN

Correlation has not yet been proven.

This distinction is important.

Player consequence for unsupported accusation

If the learner tries:

```
EVE = RESPONSIBLE
```

the system responds:

```
CLAIM NOT SUPPORTED
```

```
Known:
```

```
Eve performed an authorised earlier test.
```

```
Known:
```

```
Current activity is not part of that test.
```

```
Unknown:
```

```
Who caused the current event.
```

That is much stronger than simply marking Eve wrong.

Stage 2 outcome

The learner creates three case columns:

```
KNOWN
```

```
File 3 failed once
```

```
File 3 later changed
```

```
Credential anomaly occurred
```

```
ASSUMED
```

```
Possible actor theories
```

```
UNKNOWN
```

```
Who
```

```
How
```

```
Whether contents were viewed
```

```
Whether credential event caused transfer issue
```

This board persists throughout the incident.

STAGE 3 — TRACE THE EVIDENCE

Now we move from the timeline to the communication path.

Core model

```
SOURCE
↓
TRANSFER
↓
RECEIVER
```

The learner is not yet taught networking.

This is purely operational reasoning.

Part A — Check the source

Player opens Alice's evidence.

```
SOURCE
Alice

FILE
03_ACCESS_SCHEDULE.XLSX

ORIGINAL
AVAILABLE ✓

SOURCE CONFIRMATION
YES ✓
```

Question:

Did the problem already exist in Alice's source?

Comparison:

```
ALICE ORIGINAL
appears internally consistent
```

So:

```
SOURCE ERROR
NOT CURRENTLY SUPPORTED
```

Not impossible forever — just not supported by current evidence.

Part B — Check the transfer

The learner opens the transfer records.

```
ATTEMPT 1  
Sent from Workstation 04  
Bob: No receipt
```

```
ATTEMPT 2  
Sent from Workstation 04  
Bob: Receipt confirmed
```

Important:

We do not yet give the learner raw network packets or IPs.

They only know:

```
TRANSFER 1  
DID NOT COMPLETE AS EXPECTED  
  
TRANSFER 2  
COMPLETED BUT RESULT WAS WRONG
```

Part C — Check the receiver

Bob's destination evidence:

```
FILE RECEIVED  
03_ACCESS_SCHEDULE.XLSX  
  
CONTENT  
DOES NOT MATCH ALICE ORIGINAL
```

The learner compares the two visually.

We don't need complicated file forensics yet.

A few changed cells/records are enough.

Stage 3 key discovery

The learner can now say:

ALICE SOURCE
KNOWN

↓

TRANSFER
PROBLEM OCCURRED SOMEWHERE IN PROCESS

↓

BOB DESTINATION
DIFFERENT RESULT

Not:

The network attacker changed it.

That is still too far.

STAGE 4 — IDENTIFY WHAT FAILED

Now we introduce the security properties.

But again, **through the evidence**, not dictionary definitions first.

4A — Availability

The game asks:

Bob was authorised to receive File 3. Could he obtain it when expected?

Evidence:

ATTEMPT 1
NO

Therefore:

AVAILABILITY AFFECTED

Then give the concept:

Availability asks whether authorised users can access the information or service when required.

Now the terminology has a real experience attached to it.

4B — Integrity

Next:

When Bob finally received File 3, did it match Alice's original?

NO

Therefore:

INTEGRITY AFFECTED

Then definition:

Integrity concerns whether information remains accurate and unaltered.

Again, concept follows experience.

4C — Confidentiality

Now:

Do we know whether an unauthorised party read File 3?

The learner may be tempted to say yes.

But evidence:

PROOF SOMEONE READ CONTENTS
NONE

Therefore:

CONFIDENTIALITY — IN QUESTION

Not:

CONFIDENTIALITY FAILED

This nuance is important.

The learner understands:

A suspicious event does not prove every security property failed.

CIA board

The learner eventually builds:

CONFIDENTIALITY
IN QUESTION ?

INTEGRITY
AFFECTED ✕

AVAILABILITY
AFFECTED ✕

This is their first proper information-security assessment.

STAGE 5 — IDENTITY AND ACCESS

The incident now turns to the training credential anomaly.

Core question

Was every identity-related action expected and authorised?

The learner opens the identity activity.

Establish authorised baseline

Earlier:

EVE SECURITY CHECK

Purpose:
Credential validation

Authorised:
YES

Status:
COMPLETE

This is important because the learner saw this happen themselves.

Compare suspicious activity

Later event:

CREDENTIAL ACTIVITY

Status:

UNRECOGNISED

Authorised Eve test:

NO

Expected trainee action:

NO

The player performs:

EXPECTED?

NO

AUTHORISED?

NOT CONFIRMED

IDENTITY KNOWN?

NO

Therefore:

IDENTITY ACTIVITY — SUSPICIOUS

Not:

Identity compromised conclusively.

Again we preserve evidential discipline.

What does this introduce?

Without a long identity-security lecture, the learner experiences:

A CREDENTIAL

REPRESENTS AUTHORITY TO ACT

↓

IF CREDENTIAL ACTIVITY

DOES NOT MATCH THE EXPECTED USER



TRUST MUST BE QUESTIONED

This prepares later authentication and signature concepts.

STAGE 6 — TRUST AND VERIFICATION

This is the most important bridge into cryptography.

The learner now has several pieces:

```
FILE 3 DID NOT ARRIVE  
FILE 3 LATER ARRIVED ALTERED  
UNRECOGNISED CREDENTIAL ACTIVITY
```

But the deeper question becomes:

What can Phantom Q currently trust?

Trust board

The learner reviews:

Alice

```
ALICE  
Original source known ✓
```

Bob

```
BOB  
Receiver known ✓  
Received result verified ✓
```

Eve

```
EVE  
Authorised test known ✓  
Current suspicious activity not explained by her test  
Responsibility not established
```

Workstation / transfer process

TRANSFER PROCESS
TRUST?
NOT CURRENTLY ESTABLISHED

Information after transfer

FILE AFTER TRANSFER
TRUST?
NO

This is the key conclusion.

Verification mechanic

The learner is given a simple reasoning process to build.

The canonical algorithm we locked:

```
START
↓
CHECK SOURCE
↓
CHECK TRANSFER
↓
CHECK RECEIVER
↓
COMPARE
↓
MATCH?
  ✓      ✗
YES      NO
  ↓      ↓
VERIFY  INVESTIGATE
  ↓      ↓
STOP    STOP
```

The learner physically arranges these logic blocks.

This is an early algorithmic-thinking mechanic.

Why it matters

We are quietly teaching something broader than cybersecurity:

Do not act on a result until the process that produced it makes logical sense.

Poor algorithm design

If the learner builds:

```
START
↓
CHECK SOURCE
↓
CHECK SOURCE
↓
CHECK SOURCE
```

system:

```
LOOP DETECTED
```

```
The process does not progress
toward a conclusion.
```

If they build:

```
START
↓
COMPARE
↓
STOP
```

without checking source/receiver:

```
INSUFFICIENT EVIDENCE
```

```
The comparison cannot establish
where the mismatch occurred.
```

That makes the mechanic educational without a quiz feeling.

STAGE 7 — FINAL INCIDENT ASSESSMENT

Now the learner returns to the PQ-001 case board.

They must submit an assessment based entirely on evidence.

The locked final result is:

PQ-001 ASSESSMENT

Availability

AFFECTED

Integrity

AFFECTED

Confidentiality

IN QUESTION

Identity Activity

SUSPICIOUS

Eve Responsible

NOT CONFIRMED

Root Cause

UNKNOWN

This is critical.

Incident 01 does **not** solve the mystery.

It gives the learner a defensible statement of the problem.

5. What the learner physically does throughout Incident 01

The player should:

1. enter Training & Foundations;
2. open PQ-001;
3. arrange event cards into a timeline;
4. compare File 1/File 2 normal behaviour against File 3;
5. classify statements as fact, assumption or unknown;
6. open Alice's source evidence;
7. inspect transfer attempts;
8. open Bob's receipt evidence;
9. compare Alice and Bob's File 3 versions;
10. classify availability;
11. classify integrity;
12. assess confidentiality;

13. inspect credential activity;
14. compare authorised Eve activity with unrecognised activity;
15. assess identity status;
16. build the source → transfer → receiver verification algorithm;
17. test their algorithm;
18. submit the final PQ-001 assessment.

That is enough interaction for the incident to feel substantial.

6. Information flow

The learner should not receive everything immediately.

Beginning

We have an incident.

After reconstruction

We know the order of events.

After fact/assumption

We know which claims are supported.

After source/transfer/receiver tracing

The source and destination differ.
The transfer process cannot currently be trusted.

After CIA assessment

Availability affected.
Integrity affected.

Confidentiality unresolved.

After identity review

There is suspicious identity activity.
We still cannot identify the actor.

Final

We understand the security problem.
We still do not know the technical root cause.

That's the correct boundary.

7. Character roles in Incident 01

Trainee

Moves from:

"I saw something weird."

to:

"I can explain exactly what is known,
what failed,
and what remains unknown."

This is the first real competence progression.

Alice

Alice remains the **source-of-truth witness**.

Her role is limited to:

- confirming the original;
- answering source questions;

- validating what she actually intended.

She doesn't explain CIA.

Bob

Bob remains the **destination witness**.

He confirms:

- what did not arrive;
- what later arrived;
- what he actually saw.

Again, not a lecturer.

Eve

Eve serves as a security-reference witness.

She can show:

AUTHORISED TEST

VS

UNRECOGNISED ACTIVITY

But she is not accused.

8. World behaviour

The Headquarters should also change subtly.

Prologue end

SOC

ACTIVE

TRAINING

UNLOCKED

During Incident 01

Training & Foundations is active.

SOC remains visible but restricted.

Cryptography Lab may be visible through the HQ map but remains:

ACCESS LOCKED

This visually hints that another capability exists without telling the learner what it is yet.

9. Real concepts underneath the gameplay

Gameplay	Real concept underneath
Reorder events	Timeline reconstruction
Normal File 1/2 vs abnormal File 3	Baseline comparison
Fact/assumption/unknown	Evidence discipline
Source → transfer → receiver	Data-path reasoning
File unavailable	Availability
File changed	Integrity
No proof it was read	Confidentiality uncertainty
Credential anomaly	Identity/access risk
Compare authorised Eve test	Expected vs unauthorised activity
Build verification flow	Algorithmic reasoning
Submit PQ-001 assessment	Incident classification

Again, we keep the professional depth diluted for Release 1.

10. What we deliberately do NOT teach yet

Incident 01 should not introduce:

- encryption;

- plaintext/ciphertext;
- A7;
- public/private keys;
- IP addresses;
- routing;
- malware;
- attacker tracing;
- penetration testing;
- QKD.

The learner has identified the problem.

Now we create the need for protection.

11. Incident 01 persistent outputs

The incident adds structured assessment to the existing PQ-001 evidence.

```
PQ-001
|
├─ Prologue Evidence
|
|   ── file records
|   ── Alice source
|   ── Bob receipts
|   ── identity activity
|
└─ Incident 01 Analysis
    |
    ── reconstructed timeline
    ── fact/assumption board
    ── source-transfer-receiver trace
    ── CIA assessment
    ── identity assessment
    ── verification algorithm
```

These remain persistent.

12. Incident 01 failure logic

We should retain the same design principle as the Prologue.

Unsupported conclusion

EVE CAUSED INCIDENT

Response:

INSUFFICIENT EVIDENCE

Claiming confidentiality definitely failed

Response:

CONFIDENTIALITY FAILURE
NOT ESTABLISHED

No evidence currently confirms
that unauthorised content access occurred.

Saying availability is unaffected

System points back to:

ATTEMPT 1
BOB RECEIVED NOTHING

Not:

Wrong answer.

Saying integrity is unaffected

System compares:

ALICE ORIGINAL
≠
BOB COPY

The evidence itself challenges the decision.

13. Incident 01 success condition

The learner succeeds only when their final assessment is consistent with evidence.

Not because they identify the attacker.

Success:

```
PQ-001
INITIAL SECURITY ASSESSMENT COMPLETE ✓

FACTS
SEPARATED FROM ASSUMPTIONS ✓

AVAILABILITY
AFFECTED ✓

INTEGRITY
AFFECTED ✓

CONFIDENTIALITY
IN QUESTION ✓

IDENTITY ACTIVITY
SUSPICIOUS ✓

ACTOR
UNKNOWN ✓

ROOT CAUSE
UNKNOWN ✓
```

That is a very important kind of success:

Knowing what you do not know is part of competent security work.

14. End-state story

The trainee returns to Central Operations.

PQ-001 now displays:

```
PROBLEM UNDERSTOOD
ROOT CAUSE UNKNOWN
```

Alice still needs to send information to Bob.

But the learner has just proven:

INFORMATION
CANNOT CURRENTLY BE TRUSTED
TO MOVE SAFELY
BETWEEN AUTHORISED PARTIES

That's the locked Incident 01 closing statement.

Then Phantom Q presents a new operational requirement:

PROTECT THE INFORMATION

Cryptography Lab access becomes available.

15. Handoff into Incident 02

The transition should be extremely natural:

INCIDENT 01

We know:
the information may disappear,
change,
or potentially be exposed.

↓

QUESTION

Can Alice send information
in a form that is not immediately readable
to anyone who encounters it?

↓

CRYPTOGRAPHY LAB
UNLOCKED

↓

INCIDENT 02
PROTECT THE INFORMATION

No random jump to:

“Now learn encryption.”

The problem itself unlocks encryption.

Incident 01 master flow

```
PQ-001 OPEN
↓
ENTER FOUNDATIONS
↓
RECONSTRUCT EVENTS
↓
ESTABLISH NORMAL BASELINE
↓
SEPARATE FACT / ASSUMPTION / UNKNOWN
↓
CHECK ALICE SOURCE
↓
CHECK TRANSFER
↓
CHECK BOB RECEIVER
↓
COMPARE
↓
AVAILABILITY AFFECTED
↓
INTEGRITY AFFECTED
↓
CONFIDENTIALITY IN QUESTION
↓
INSPECT CREDENTIAL ACTIVITY
↓
COMPARE AUTHORISED / UNRECOGNISED ACTION
↓
IDENTITY ACTIVITY SUSPICIOUS
↓
BUILD VERIFICATION ALGORITHM
↓
TEST REASONING
↓
SUBMIT PQ-001 ASSESSMENT
↓
EVE RESPONSIBLE? NOT CONFIRMED
↓
ROOT CAUSE? UNKNOWN
↓
INFORMATION CANNOT CURRENTLY
BE TRUSTED TO MOVE SAFELY
↓
CRYPTOGRAPHY LAB UNLOCKED
↓
INCIDENT 02
```

This gives us a complete **Incident 01 design specification** while preserving the original seven-stage structure we agreed on.